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Effects of Different Set Configurations on Barbell Velocity and Displacement During a Clean Pull

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ABSTRACT

The effects of 3 types of set configurations (cluster, traditional, and undulating) on barbell kinematics were investigated in the present study. Thirteen men (track and field = 8; Olympic weightlifters = 5) (mean ± *SEM* age, 23.4 ± 1.1 years; height, 181.3 ± 2.1 cm; body mass, 89.8 ± 4.2 kg) performed 1 set of 5 repetitions in a cluster, traditional, and undulating fashion at 90 and 120% of their 1 repetition maximum (1RM) power clean (119.0 ± 4.3 kg). All data were collected at 50 Hz and analyzed with a V-Scope Weightlifting Analysis System. Peak velocity (PV) and peak displacement (PD) were analyzed for each repetition and averaged for each set type. Results indicated that a significantly ($p < 0.016$) higher PV occurred during the cluster set when compared with the traditional sets at both intensities. PD was significantly higher than traditional sets at the 120% intensity. The present study suggests set configuration can affect PV and PD during clean pulls.

Key Words: weightlifting, kinematics, V-scope, cluster set, Olympic weightlifting

Reference Data: Haff, G.G., A. Whitley, L.B. McCoy, H.S. O'Bryant, J.L. Kilgore, E.E. Haff, K. Pierce, and M.H. Stone. Effects of different set configurations on barbell velocity and displacement during a clean pull. *J. Strength Cond. Res. 17(1):95–103. 2003.*

Introduction

In the sport of Olympic weightlifting, the pull refers to the portion of the snatch and clean in which the barbell is displaced from the floor to approximately waist height (4). There are several critical factors that may affect the contribution of the pull to performance in the snatch and clean. It has been suggested that the trajectory (24), velocity (9, 10), and displacement (4, 9) of the barbell are closely related to lifting success. Additionally, some authors have suggested that the manner in which the forces are applied to the barbell are related to lifting success (11).

Garhammer and Hatfield (10) suggest that the most important factor related to the pull during the snatch and clean is that the barbell's final velocity must allow the lifter time to move into the catch position, where the lifter receives the barbell. Several authors have reported pulling velocities between 1.14 and 1.86 $\text{min}\cdot\text{s}^{-1}$ when examining elite weightlifters in actual competition (8, 10, 16, 18). Bartonietz (3) suggests that greater vertical barbell velocities will result in greater vertical barbell displacements during the pull, which may potentially improve lifting performance.

Enoka (4) suggests that the actual barbell displacement during the snatch and clean are critical during weightlifting. The vertical barbell displacement achieved during the pull portion of the clean appears to fall between 99.6 ± 7.0 and 106.2 ± 8.5 cm in elite weightlifters competing in competition (8, 16, 18). When looking exclusively at the clean pull exercise performed at 80% of a 1 repetition maximum (RM) clean, the barbell displacement of 6 elite American weightlifters has been reported to be approximately 102.3 cm (8). Interestingly, the barbell displacement during an actual clean with the same resistance was approximately 95 cm in the same athletes (8). It is important to note that the barbell displacement may also be a function of the lifter's height, with larger athletes having to pull the barbell through a greater displacement in order to achieve a successful lift. The maximal height achieved during the pull phase of a squat clean has been reported to be 60–61% of the athlete's height.

The maximal height and barbell velocity achieved during the pull portion of a clean appears to be of critical importance to lifting success. Thus selecting exercises, which maximize pulling characteristics, may ultimately lead to improved lifting performance. Tra-

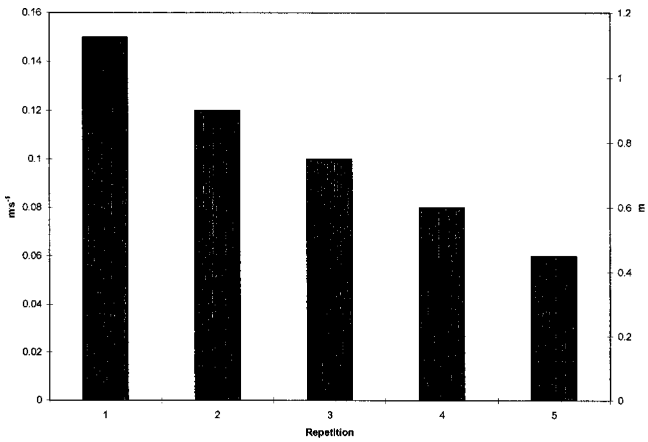

Figure 1. Theoretical velocity and displacement model for a traditional set.

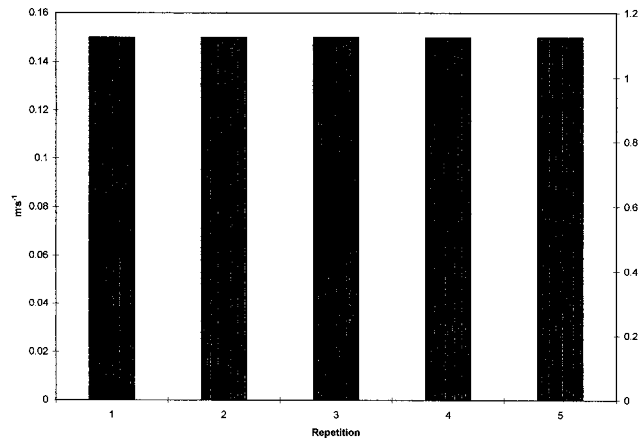

Figure 2. Theoretical velocity and displacement model for a cluster set.

ditionally, snatch and clean pulls are employed as assistance exercises by Olympic weightlifters (15) and athletes in other sports. In this context, these lifts are often performed for multiple repetitions at intensities that range from 70 to 120% of the athletes' maximum clean or snatch (1). The incorporation of pulling motions is often done for all levels of lifters in an attempt to improve the athletes' pulling strength, ability to generate high velocities, and ability to maximize vertical barbell displacements (24). Therefore, any manipulations in the training set configuration that enhances the ability of these exercises to generate higher velocities or greater vertical barbell displacements may be advantageous to weightlifters and strength power athletes at any level.

There are several ways in which the athlete can manipulate the training set configuration. Traditionally, the configuration of a training set is composed of each repetition being performed in a continuous fashion without rest between repetitions (5, 21). For the present study, this type of set configuration was termed a traditional set. Another training set configuration that has been utilized classically by strength athletes is the rest-pause set (5) or the cluster set (20). This type of set involves performing a training set with 10–30 seconds rest between repetitions with near-maximal weights (5, 20). For the present study, this type of set configuration will be termed a cluster set. The final type of set configuration that we decided to examine involved manipulating the resistance in a pyramid-type fashion during a cluster-type set. This set was designed in an attempt to induce a contrast loading effect when the resistance was reduced during the last 2 repetitions of the set (2). For the present study we termed this set configuration as an undulating set.

It is our hypothesis that when performing multiple repetitions of clean pulls in a traditional manner, the barbell velocity and displacement (Figure 1) will progressively decrease as muscular fatigue manifests itself. We suggest that performing a cluster set with 15–30 seconds rest between repetitions will maximize barbell velocity and displacement (Figure 2) for each individual repetition during a set. Additionally, we suggest that the undulating set will produce a distinctly different velocity and displacement pattern as a result of contrast loading (2) (Figure 3). Therefore, the purpose of this investigation was to examine the effects of manipulating set configurations on barbell velocity and vertical displacement.

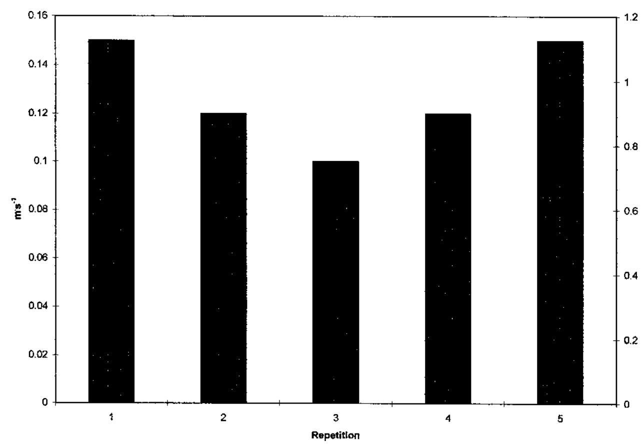

Figure 3. Theoretical velocity and displacement model for an undulating set.

Methods

Experimental Aproach to the Problem

All subjects participated in 2 testing sessions separated by 1 week. Each subject was required to participate in the investigation for a total of 11 days. For 2 days prior to each testing session, all subjects were asked to refrain from physical activity that might be detrimental to performance. On day 0, each subject completed all paperwork related to University Ethical Review Committee guidelines and contraindication screening. Testing during this session was utilized to determine the subject's biometric data and 1 repetition maximum

Measurements:

• = Pull Study Protocol
♦ = 1-RM Power Clean
⊗ = Biometric Data
∇ = Refrain from physical activity prior to testing
⊕ = Regular Strength Training Days

Days										
-2	-1	0	1	2	3	4	5	6	7	8
∇	∇	∇ ⊗ ♦		⊕	⊕		⊕	∇	∇	∇ •

Figure 4. Experimental design.

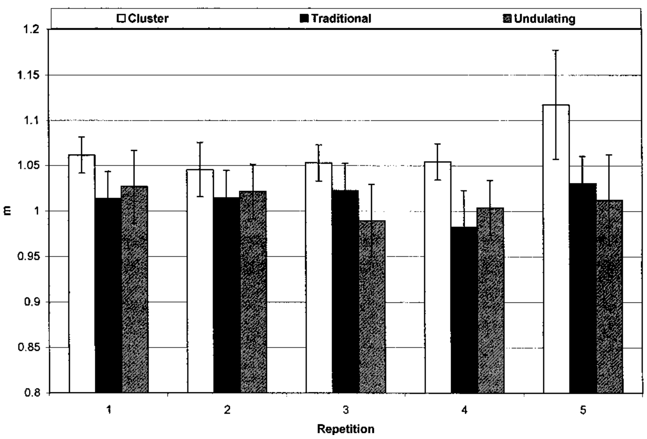

Figure 5. Barbell displacement during multiple set configurations at 90% of 1RM.

(1RM) in the power clean. Subjects were then instructed that they could perform their regularly scheduled workouts on days 2, 3, and 5. Subjects were also asked to refrain from physical activity on days 6 and 7. On day 8 subjects reported to the lab to participate in pull testing. The pull testing session required the subjects to do pulls with 90 and 120% of their 1RM power clean for undulating, cluster, and traditional set configurations. To account for a treatment order effect these intensities and set configurations were randomly assigned in a counterbalanced fashion. The use of this testing model allowed for a detailed comparison of barbell kinematics at the different intensities and set configurations. Figure 4 presents the experimental design in detail.

Subjects

Thirteen men (track and field = 8; Olympic weightlifters = 5) who were currently performing power cleans and clean pulls in their training participated in this investigation. All track and field athletes and Olympic weightlifters analyzed in this study had qualified for the Collegiate National Weightlifting Championships. The 8 track and field athletes all competed in Olympic weightlifting in the off-season but were classified as track athletes because track and field was their primary sport. Written informed consent was received from all subjects in accordance with University Ethical Review Committee guidelines. All subjects were screened for contraindications to exercise with the use of a training history and medical/health history questionnaire. Subject characteristics are presented in Table 1.

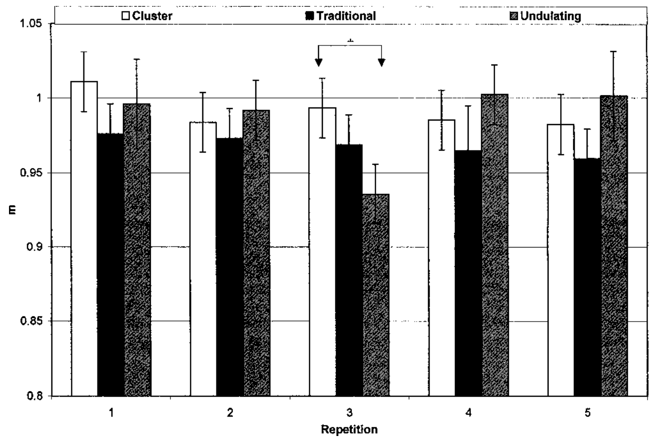

Figure 6. Barbell displacement during multiple set configurations at 120% of 1RM. * = significant difference $p < 0.02$.

Table 1. Subject characteristics.

Variable	Mean ± *SEM*
Age (y)	23.4 ± 1.1
Height (cm)	181.3 ± 2.1
Body mass (kg)	89.9 ± 4.2
Body fat (%)	13.3 ± 2.1
1RM power clean (kg)	119.0 ± 4.3
Training experience (y)	3.4 ± 0.9

Preliminary Testing (Day 0)

During this testing session the subjects had their height, body mass, and body composition determined. Height and body mass were determined using a wall-mounted stadiometer and a Healthometer physician's scale (Continental Scale Corp., Bridgeview, IL). Body composition was determined using Lange skinfold calipers (Cambridge Scientific Industries, Cambridge, MD) and a 7-site skinfolds measurement (14). Test-retest reliability was determined to be $R = 0.99$ for the skinfold procedures. Subjects were then tested for their 1RM power clean with previously established methods (21). The 1RM power clean was then used to calculate the 90 and 120% intensities tested in this study. Typically, in the sport of weightlifting the intensity of pulling motions is calculated from a maximal lift (17, 22), such as the snatch, clean, power snatch, or power clean. Typically, weightlifters utilize intensities in the 70–120% range of the athletes' maximum during certain phases of training (1). The 90% intensity was selected for this study because this intensity has been suggested to be optimal for pulling motions (7, 17). Pulls with higher intensities (120%) are also performed by weightlifters during certain phases of training (17). However, it has been suggested that pulls performed at this intensity may result in deviations in kinematic parameters such as velocity and barbell displacement (17). Therefore, an intensity of 120% was selected for the present study in an attempt to see if specific kinematic parameters could be optimized with manipulations in set configuration.

Primary Testing (Day 8)

All subjects reported to testing after participating in a mandatory abstinence from physical activity (days 6 and 7). On day 8 all subjects performed 1 set of 5 repetitions in a cluster, traditional, or undulating fashion at both the 90 and 120% intensities. The set configuration and testing intensities were randomly assigned. If subjects were assigned the 90% intensity first they performed 5 minutes of brief jogging followed by a weightlifting specific warm-up procedure: 5 repetitions at 45 and 70% of their 1RM power clean. If the 120% intensity was selected as the first trial, subjects again performed 5 minutes of brief jogging followed by a weightlifting specific warm-up procedure: 5 repetitions at 45 and 70% of their 1RM power clean, 3 repetitions at 95% of their 1RM power clean, and 1 repetition at 105% of their 1RM power clean. Subjects were given 3 minutes rest between sets (warm-up and target) at each intensity.

Instrumentation

All pulling data were collected with a V-Scope Weightlifting Analysis System (Lipman Electronic Engineering Ltd., Ramat Hahayal, Israel). The V-Scope was set up 3 m from the barbell and utilized a 50-Hz sampling rate. Data were collected on the following variables: peak vertical displacement (PD), peak power (PP), peak velocity (PV), peak acceleration (PA), time to PD, time to PP, time to PV, and time to PA. The reliability and validity of the V-Scope has been previously tested in our laboratory. Displacement reliability and validity was checked by moving ($n = 100$) the ultrasound-emitting button both vertically and horizontally through a known distance. No significant differences were found between actual and measured horizontal and vertical displacements. Additionally, displacement reliability was found to be very high ($R = 0.97$). The reliability for PP ($R = 0.82$), PV ($R = 0.79$), PF ($R = 0.82$), time to PP ($R = 0.70$), and time to PV ($R = 0.81$) have also been examined through test-retest trials ($n = 220$). Data were collected for each repetition of each set, and average values were also calculated for each condition by averaging the data over the 5 repetitions.

Statistical Analyses

A 3×5 repeated measures (set configuration $\times$ repetition) analysis of variance (ANOVA) was used to analyze all variables at each intensity. Average values were calculated for each variable collected during each set were analyzed with a 2×3 repeated measures (intensity $\times$ set configuration) ANOVA. Significance was set at $p \leq 0.05$ for all ANOVA tests. When significant F values were determined, paired-tests were used, in conjunction with Holm's sequential Bonferroni method for controlling type I error, to determine the significant differences (13). All data are reported as means $\pm$ *SEM*. All statistics were performed with SPSS 10.0 (SPSS Inc., Chicago, IL).

Results

The effects of set configurations at 90, and 120% of a 1RM power clean were examined during this investigation. Individual repetition characteristics are listed in Tables 2 and 3. The effects of the different set configurations on the 5 repetition averages are presented in Table 4. When examining the 5 repetition averages during each set configuration and intensity condition, it was determined that the cluster set exhibited significantly (90%: $p = 0.007$; 120%: $p = 0.009$) higher barbell velocities than the traditional set. There were no differences in barbell velocities between the undulating set and the cluster or traditional set at the 90 or 120% intensity. All velocities achieved in each set configuration during the 90% trials were significantly higher than those achieved during the 120% intensity trials. Additionally, the performance of the cluster set resulted in significantly higher ($p = 0.01$) barbell displacements when compared with the traditional set at the 120% intensity. There were no differences between the displacements achieved during any of the set configurations during the 90% intensity trials (Figure 5). It is important to note that the difference between the cluster and traditional set displacements during the 90% trial approached statistical significance ($p = 0.02$). There were no significant differences in barbell displacement between the 90 and 120% intensities for the traditional and undulating set configurations (Figure 6). The cluster set resulted in a significantly ($p = 0.001$) higher barbell displacement during the 120% trials. All barbell displacements achieved during the 90% condition were significantly higher than the 120% condition. Additionally, there was a significant difference between the percentage of body height of the pull displacement between the cluster set (54.7 $\pm$ 0.01%) and traditional set (53.4 $\pm$ 0.01%) ($p = 0.011$), and the undulating set (54.4 $\pm$ 0.01%) and traditional set (54.7 $\pm$ 0.01%) ($p = 0.08$) during the 120% condition. There were no differences between the percentage of body height achieved during the cluster set (58.8 $\pm$ 0.01%), traditional set (55.9 $\pm$ 0.02%), and undulating set (55.8 $\pm$ 0.02%) performed during the 90% condition. No differences in PP, time to PP, PA, time to PA, time to PV, and time to PD during either condition were observed. However, the 120% condition produced significantly longer time to PP, time to PA, time to PV, and time to PD than the 90% intensity.

Table 2. Clean pull barbell kinematics for 90% intensity (mean ± *SEM*).*

		90% Repetitions				
Variable	Set	1	2	3	4	5
PP (W)	TR	2,768.33 ± 214.99	2,830.54 ± 179.84	2,879.88 ± 283.93	2,762.31 ± 237.37	2,599.08 ± 189.37
	C	2,889.85 ± 206.75	2,905.88 ± 235.46	3,083.65 ± 232.99	2,949.13 ± 221.03	2,955.82 ± 241.15
	UND	2,683.09 ± 242.85	2,776.03 ± 268.21	2,857.32 ± 257.95	2,598.77 ± 173.00	3,060.57 ± 306.74
Time to PP (ms)	TR	0.76 ± 0.05	0.74 ± 0.04	0.74 ± 0.03	0.74 ± 0.05	0.73 ± 0.04
	C	0.77 ± 0.04	0.79 ± 0.04	0.77 ± 0.04	0.75 ± 0.04	0.79 ± 0.05
	UND	0.74 ± 0.05	0.80 ± 0.04	0.82 ± 0.05	0.77 ± 0.05	0.73 ± 0.04
PA ($m \cdot s^{-1}$)	TR	9.41 ± 1.08	9.17 ± 0.96	9.75 ± 1.16	9.00 ± 1.05	8.83 ± 0.95
	C	9.61 ± 0.90	9.21 ± 0.72	10.27 ± 0.92	9.75 ± 0.85	9.72 ± 0.94
	UND	8.45 ± 0.97	9.07 ± 1.30	8.54 ± 1.05	9.06 ± 1.13	10.79 ± 1.21
Time to PA (ms)	TR	0.74 ± 0.05	0.72 ± 0.04	0.73 ± 0.03	0.72 ± 0.04	0.71 ± 0.04
	C	0.76 ± 0.04	0.77 ± 0.04	0.75 ± 0.04	0.72 ± 0.04	0.77 ± 0.05
	UND	0.72 ± 0.05	0.78 ± 0.04	0.80 ± 0.04	0.75 ± 0.05	0.72 ± 0.04
Time to PV (ms)	TR	0.80 ± 0.05	0.78 ± 0.04	0.79 ± 0.04	0.78 ± 0.04	0.77 ± 0.04
	C	0.82 ± 0.04	0.83 ± 0.04	0.83 ± 0.04	0.87 ± 0.04	0.84 ± 0.05
	UND	0.80 ± 0.05	0.83 ± 0.04	0.86 ± 0.04	0.82 ± 0.04	0.78 ± 0.04
Time to PD (ms)	TR	0.97 ± 0.04	0.95 ± 0.03	0.97 ± 0.03	0.97 ± 0.04	0.96 ± 0.03
	C	0.99 ± 0.04	0.99 ± 0.04	0.98 ± 0.04	0.98 ± 0.04	1.02 ± 0.05
	UND	0.95 ± 0.05	1.01 ± 0.04	1.03 ± 0.05	0.98 ± 0.05	0.95 ± 0.04

* PP = peak power; PV = peak velocity; PD = peak displacement; PA = peak acceleration; TR = traditional; C = cluster; UND = undulating.

Table 3. Clean pull barbell kinematics for 120% intensity (mean ± *SEM*).*

Variable	Set	120% Repetitions 1	2	3	4	5
PP (W)	TR	2,953.87 ± 233.15†	2,738.24 ± 302.56	2,894.18 ± 270.61	2,574.74 ± 224.66	2,316.32 ± 234.72
	C	3,124.58 ± 198.20	2,762.20 ± 202.23	2,762.15 ± 220.04	2,614.57 ± 254.28	2,648.05 ± 237.25
	UND	2,895.11 ± 253.07	2,928.42 ± 238.71‡	2,440.15 ± 258.45	2,722.32 ± 245.32	2,482.11 ± 302.63
Time to PP (ms)	TR	1.01 ± 0.05	0.95 ± 0.04	0.98 ± 0.05	0.97 ± 0.06	0.97 ± 0.06
	C	0.98 ± 0.05	1.02 ± 0.05	0.99 ± 0.06	1.04 ± 0.05§	1.04 ± 0.06
	UND	0.88 ± 0.05	0.99 ± 0.07‖	1.18 ± 0.06¶,#	1.04 ± 0.06§	0.91 ± 0.05
PA ($m \cdot s^{-1}$)	TR	7.92 ± 0.81	7.82 ± 0.85	8.77 ± 0.94¶	7.10 ± 0.99	6.49 ± 0.81
	C	8.32 ± 0.73	8.62 ± 0.89	7.71 ± 0.63	6.78 ± 0.76	6.85 ± 0.83
	UND	8.47 ± 0.90	7.69 ± 0.82	6.88 ± 0.85	7.41 ± 0.91	6.88 ± 1.11
Time to PA (ms)	TR	0.99 ± 0.05	0.92 ± 0.04	0.96 ± 0.05	0.95 ± 0.06	0.93 ± 0.06
	C	0.95 ± 0.05	0.99 ± 0.05	0.96 ± 0.06	1.00 ± 0.05	1.00 ± 0.06
	UND	0.85 ± 0.06	0.96 ± 0.07‖,**	1.15 ± 0.06¶,#	1.10 ± 0.11	0.87 ± 0.06
Time to PV (ms)	TR	1.05 ± 0.05	0.99 ± 0.04	1.02 ± 0.05	1.03 ± 0.06	1.03 ± 0.05
	C	1.01 ± 0.05	1.06 ± 0.05	1.03 ± 0.06	1.08 ± 0.05	1.08 ± 0.06
	UND	0.94 ± 0.05	1.04 ± 0.06	1.22 ± 0.06	1.08 ± 0.06	0.78 ± 0.04
Time to PD (ms)	TR	1.20 ± 0.06	1.16 ± 0.04	1.19 ± 0.05	1.23 ± 0.05	1.19 ± 0.05
	C	1.18 ± 0.05	1.22 ± 0.06	1.17 ± 0.06	1.24 ± 0.05	1.26 ± 0.06
	UND	1.10 ± 0.05	1.20 ± 0.06	1.37 ± 0.07#	1.25 ± 0.06§	1.07 ± 0.03

* PP = peak power; PV = peak velocity; PD = peak displacement; TPP = time to peak power; PA = peak acceleration; TR = traditional; C = cluster; UND = undulating.
† repetition 1 > repetition 5.
‡ repetition 2 > repetition 3.
§ repetition 4 > repetition 1.
‖ repetition 2 > repetition 1.
¶ repetition 3 > repetition 5.
# repetition 3 > repetition 1.
** repetition 2 > repetition 5.

Table 4. Performance characteristics: five repetition averages (mean ± *SEM*).*

Variables	Condition					
	90%			120%		
	C	TR	UND	C	TR	UND
PP (W)	2,956.87 ± 190.58	2,768.03 ± 199.71	2,795.15 ± 224.64	2,782.31 ± 203.69	2,795.47 ± 225.72	2,693.62 ± 242.73
PV ($m \cdot s^{-1}$)	1.72 ± 0.06†‡	1.59 ± 0.06‡	1.63 ± 0.10‡	1.37 ± 0.05†	1.27 ± 0.05	1.32 ± 0.05
PD (m)	1.07 ± 0.03‡	1.01 ± 0.03‡	1.01 ± 0.04‡	0.99 ± 0.02†	0.97 ± 0.02	0.99 ± 0.02§
PA ($m \cdot s^{-1}$)	9.71 ± 0.76‡	9.23 ± 0.95‡	9.18 ± 0.97‡	7.66 ± 0.62	7.62 ± 0.77	7.47 ± 0.85
Time to PP (ms)	0.77 ± 0.04	0.74 ± 0.03	0.77 ± 0.04	1.01 ± 0.05‖	0.98 ± 0.05‖	1.00 ± 0.04‖
Time to PA (ms)	0.75 ± 0.04	0.72 ± 0.03	0.75 ± 0.04	0.98 ± 0.05‖	0.95 ± 0.05‖	0.97 ± 0.06‖
Time to PV (ms)	0.84 ± 0.04	0.78 ± 0.03	0.82 ± 0.04	1.05 ± 0.05‖	1.02 ± 0.05‖	1.05 ± 0.05‖
Time to PD (ms)	0.99 ± 0.04	0.96 ± 0.03	0.98 ± 0.04	1.21 ± 0.05‖	1.19 ± 0.05‖	1.23 ± 0.05‖

* PP = peak power; PV = peak velocity; PD = peak displacement; C = cluster; TR = traditional; UND = undulating.
† Cluster > traditional.
‡ 90% > 120% ($p < 0.05$).
§ Undulating > traditional.
‖ 120% > 90% ($p < 0.01$).

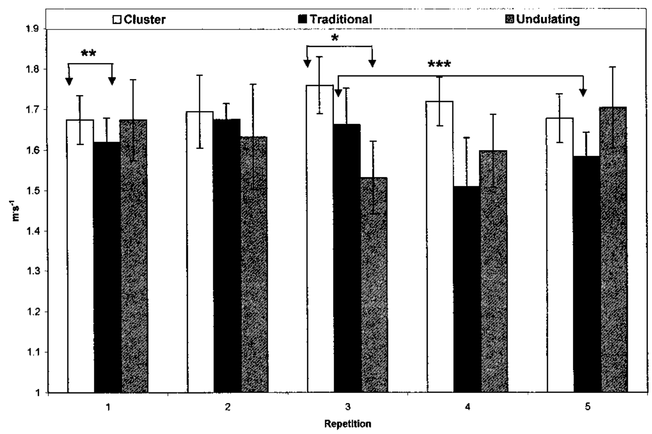

Figure 7. Barbell velocity during multiple set configurations at 90% of 1RM. * = significant difference $p < 0.02$. ** = significant difference $p < 0.03$. *** = significant difference $p < 0.01$.

Discussion

This investigation aimed to determine whether manipulating set configurations stimulated the occurrence of the hypothetical models (Figures 1–3) for barbell velocity and displacement. These models suggest that barbell velocity and displacement will decrease during the performance of a traditional set configuration and that the cluster set configuration will allow for the maintenance of these 2 performance markers. The results of this investigation suggest that the hypothetical model is in fact a valid depiction of the changes in barbell velocity and displacement during traditional, cluster, and undulating sets.

The traditional set configuration resulted in an overall trend, which suggested that barbell velocity declined with each repetition during the performance of this type of set. This can be seen when examining the graphical representation of the data (Figures 7 and 8). The observation of this trend may have implications for the competitive weightlifter and many other strength power athletes. Several researchers suggest that the ability to generate high barbell velocities during Olympic weightlifting movements is closely related to lifting success (9, 10). The athletes studied in this investigation generated an average velocity of 1.59 ± 0.06 $min \cdot s^{-1}$ during the 90% trial and 1.27 ± 0.05 $min \cdot s^{-1}$ during the 120% trial while performing a clean pull with a traditional set configuration. In comparison, the average barbell velocity was significantly higher in the cluster set at both the 90 and 120% intensities (90%, 1.72 ± 0.06 $min \cdot s^{-1}$; 120%, 1.37 ± 0.05 $min \cdot s^{-1}$). The fact that the cluster set configuration resulted in significantly higher barbell velocities may be related to the fact that partial recovery may have been stimulated in the 30 seconds rest interval that was used between repetitions. Hakkinen (12) has reported that strenuous heavy resistance training can result in

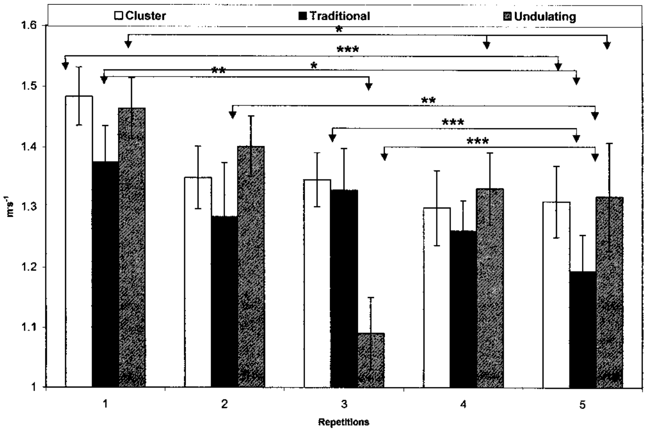

Figure 8. Barbell velocity during multiple set configurations at 120% of 1RM. * = significant difference $p < 0.02$. ** = significant difference $p < 0.001$. *** = significant difference $p < 0.01$.

acute fatigue in the neuromuscular system, which may manifest itself as a decrease in force production and in voluntary neural activation of the exercised muscle.

Decreases in maximal force generating capabilities, rate of force development, and rate of relaxation may occur in as little as 5 to 9 maximal contractions (23). These alterations in maximal force production capabilities and force-time curve characteristics may be related to increases in blood lactate concentrations (23). It is possible that the composition of the cluster set configuration, with its 30 seconds of extra recovery time between repetitions, allowed for some replenishment of phosphocreatine (PCr) stores, while the traditional set configuration resulted in a greater depletion of these stores and ultimately resulted in increased lactate production. Sahlin and Ren (19) suggest that adenosine triphosphate (ATP) and PCr concentrations are significantly decreased after maximal voluntary contractions. These decreases in ATP and PCr occur simultaneously with significant elevations in lactate concentrations and significant decreases in the amount of force the muscle can generate. In this study it was reported that after 15 seconds of recovery the ability to perform a maximal voluntary contraction was restored to 79.7 ± 2.3% of the initial force value (19). Additionally, Wootton and Williams (25) have reported that the inclusion of 30 extra seconds of recovery can significantly elevate power output generating capabilities over 5 maximal cycle ergometer sprint bouts. It was suggested that the differences reported in maximal sprint performance were related to the marked increase in blood lactate seen in the bout of exercise, which contained the short rest interval (25). Elevations in blood lactate may have a negative effect on muscle contraction (19). This possible negative effect may result in changes in the contractile characteristics of the muscle as a result of impairments in the ATP generating process (19). Even though blood lactate concentrations were not measured in the present investigations, it is possible that the traditional set configuration resulted in a greater production of lactate than the cluster set configuration, thus partially explaining the differences depicted in barbell velocity seen with each type of set structure. Therefore, it is possible that the traditional group experienced greater fatigue as a result of a pronounced reduction of PCr stores than the cluster set group. Based on the work of Sahlin and Ren (19) it is likely that the cluster set group was able to have a very rapid recovery of force-generating capabilities that resulted in improved pulling performance. This possibility may partially explain the differences seen between the set configurations and suggest that the cluster set configuration may be useful when attempting to maximize barbell velocity during a training session or bout.

It has been suggested that the maximal barbell velocity during the pull phase is strongly related to the vertical displacement of the barbell (3). Therefore it might be hypothesized that the cluster set configuration would have higher barbell displacements than the traditional set. However, this hypothesis only held true for the 120% intensity with significant differences only occurring during this condition. It is important to note that the difference between the cluster set and traditional set barbell displacement approached significance. Based on these findings it appears that decreases in barbell velocity occur along with changes in barbell displacement. It is possible that the same mechanisms of fatigue that may have affected the barbell velocity resulted in a decreased ability to maximize the barbells' vertical displacement during the clean pulls performed at both 90 and 120% of the maximal power clean.

Based on the concept of velocity and movement specificity it appears that the cluster set configuration allows the athlete to optimize pull velocity and displacement. Continual use of the cluster set may result in the development of weightlifting specific strength endurance over time. The contention that weightlifting specific endurance is improved with the implementation of a cluster configuration centers around fact that the rest interval between repetitions of the set allows for partial recovery by the athlete (20). It has been noted in the literature that 15 seconds of recovery can result in a 79.7 ± 2.3% restoration of maximal force-generating capabilities after a maximal voluntary contraction. This ability to partially recover between repetitions allows the athlete to handle near maximal weights for many repetitions. The ability to elevate both volume and intensity (volume load) will most likely result in elevations in hypertrophic and neural adaptations as a result of the training bout (6). However, further research is needed to determine the effectiveness of the cluster set as a key component of a periodized program.

One of the classic arguments against doing sets with high repetition schemes centers on the fact that weightlifting technique becomes impaired with increasing fatigue. Therefore, the vast majority of competitive weightlifters perform sets with repetition schemes that range between 1 and 5 repetitions per set depending upon the phase of training and type of exercise employed in the training session (15). Based on the data presented in the present study the cluster set configuration may allow weightlifters and other strength athletes to perform pulls with higher intensities for higher volumes, while maintaining barbell velocities and displacements. Theoretically, the performance benefits of training programs, which utilize the cluster set configuration, should result in improvements in lifting performance. Further research is needed to completely understand the role of the cluster set in improving lifting performance.

Practical Applications

When designing resistance-training programs that impact distinctive performance characteristics many factors need to be considered. In the present study factors that affect barbell velocity and displacement were investigated. In this investigation it appears that configuring the set schemes of a resistance-training program with a cluster set model may produce specific alterations to both performance parameters. These alterations to both the barbell velocity and displacement may ultimately result in improved performance due to the relationship of these variables to lifting performance. Even though the present study examined an acute effect of different set schemes on barbell velocity, the data suggest that the cluster set configuration may allow the athlete to perform greater volumes of work while not experiencing the impairments in barbell kinematics normally associated with high volume work. However, it is important to note that more research is needed in order to fully understand the role of the cluster set in the training practices of weightlifters and other strength power athletes.

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